MORE ABOUT THE 737 MAX & ENGINEERING ETHICS

ENP-0112

2/23/2026

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Congress Passes “Aircraft Certification, Safety, and Accountability Act”

By **Marc Cook** - Published: December 22, 2020

Boeing & Aerospace

The Seattle Times

Congress clears Boeing 737 MAX 7 and 10 jets, adding a condition

Dec. 19, 2022 at 11:45 pm | *Updated Dec. 23, 2022 at 7:51 pm*

A BIT MORE CONTEXT

- Lion Air 43 was the previous 96-min flight in accident aircraft.
  - The stick shaker was active for the entire flight, but the crew shut off the stabilizer trim, deactivating MCAS. The crew reported some malfunctions, but did not specify the stick shaker activation or stabilizer trim cutout.
- Lion Air was the first of two similar accidents. The second was Ethiopian 302 in March 2019.
  - An advisory bulletin had been circulated in November 2018, but it is not clear whether the accident crew saw or understood the bulletin.
  - In a rare step, the NTSB and French authority (BEA) both published comments immediately after the final report of ET302 was published, disputing some findings in the final EAIB report.

Chandra, D.C., Sparko, A.L. Kendra, A. (2023). Flightcrew Response to Aircraft System Failures, Malfunctions, and Systems Not Functioning as Expected. DOT-VNTSC-FAA-23-05. U.S. DOT Volpe National Transportation Systems Center. https://rosap.ntl.bts.gov/view/dot/72503

MORE ON BOEING 737 MAX ACCIDENTS

- https://www.faa.gov/foia/electronic_reading_room/boeing_reading_room
- For example:
  - A **Technical Advisory Board (TAB)** was formed, consisting of an independent team of aircraft safety, system, and flight test experts. The TAB provided an independent technical evaluation of the B737 MAX MCAS and its redesign to inform the FAA decision on MCAS approval and return to service.
  - The **Joint Authorities Technical Review (JATR)** reviewed the type certification of the flight control system on the B737 MAX. The FAA chartered the JATR to review the work conducted during the B737 MAX certification program, to assess whether compliance was shown with the required applicable airworthiness standards related to the flight control system and its interfaces, and to recommend improvements to the certification process if warranted. The JATR team's review also focused on flight crew training and operational suitability of the design.
- See also, NTSB recommendations from full Lion Air Report
  - On NTSB website and in Canvas, under Readings - Supplementary

ONE OUTCOME (OUT OF MANY)
FAILURE EFFECT CLASSIFICATIONS UPDATED

§ 25.1309 Equipment, systems, and installations.

Excerpt

(b) The airplane systems and associated components, evaluated separately and in relation to other systems, must be designed and installed so that they meet all of the following requirements:

(1) Each catastrophic failure condition—

(i) Must be extremely improbable; and

(ii) Must not result from a single failure.

(2) Each hazardous failure condition must be extremely remote.

(3) Each major failure condition must be remote.

AC 25.1309-1B (8/30/2024) provides details on one means of compliance.

(This replaced version A, which was in effect from 1988 until the 2024 version.)

MITIGATIONS RELATED TO 737MAX ACCIDENTS

Individual

- Selection and training (Knowledge, Skills, and Abilities)
- Motivation, arousal (optimize attention)

Checklists & other simulator training

Organization

- Policies and procedures (e.g., checklists)
- Safety culture

Airline (operator) & Maintenance
Original Equipment manufacturer
Regulator

System Design

Sensors and Alerts, Automated-system design

- Controls (soft or hard), "Usability" (e.g., visibility of system state, interaction with data)
- Automated systems? (take the operator out of the system)

Operational Risk

- Variation across each operation (e.g., across flights)
- Work as imagined (theory) vs. work as done (reality)

VOLPE REPORT

What was the *pilot response* to the system design?

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Image from Volpe Annual Review, 2025
https://rosap.ntl.bts.gov/view/dot/79090

How do Flightcrews Respond to Aircraft Systems Failures?

ANALYZING OPERATIONAL DATA SHOWS HOW PILOTS HANDLE REAL SITUATIONS

FAA asked the **U.S. DOT Volpe Center** to study the pilot response of five air carrier accidents with major aircraft system failures. This work will help to integrate human factors findings into FAA regulatory processes and materials. One task was to conduct a comprehensive review of literature about:

- AIRCRAFT SYSTEM PROBLEMS,
- ALERTS AND CHECKLISTS,
- PILOT TRAINING, AND
- PILOT RESPONSE.

The Volpe Center Aviation Human Factors Team analyzed **five public accident reports** and 20 records from the **NASA Aviation Safety Reporting System** (ASRS) public database.

The U.S. DOT Volpe Center's Transportation **Human Factors team** prepared a final report that can inform **training, policy,** and **procedures** for air carrier training.

Flightcrew Response to Aircraft System Failures, Malfunctions, and Systems Not Functioning as Expected

U.S. Department of Transportation Volpe Center

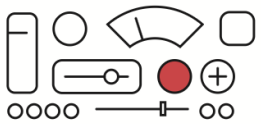
- The team developed a method for extracting and summarizing data about pilot response from operational reports and explored the limits of this method.
- They partitioned the research space (e.g., training, alerts, failure types) to understand the connections between unexpected system behaviors and pilot response.

WHAT ARE SOME KEY INSIGHTS FROM THE ACCIDENT ANALYSES?

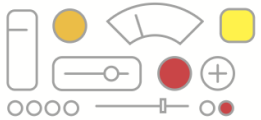
- Pilots are regularly exposed to aircraft system failures, malfunctions, and **unexpected behavior**.
- Many technological problems can be addressed, but **human behavior is much more difficult to change**.
- **Pilot experience** with unexpected situations in modern aircraft varies and will continue to evolve as the pilot workforce changes.

FIVE KEY OBSERVATIONS FROM THE ASRS RECORDS:

1. Determining **whether or not a malfunction has actually occurred** can be challenging.



2. Diagnosing malfunctions can be difficult, **especially when there are conflicting indications** or intermittent malfunctions.



3. **Checklists** and standard operating procedures are often helpful.

4. However, checklists may be **insufficient** for diagnosing and resolving **multiple malfunctions**.

5. It is challenging for pilots to decide when to **stop troubleshooting the malfunction** and **begin planning to divert**.

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INSIGHTS FROM VOLPE REPORT

- Pilot experience impacts effective management of the situation.
  - Pilot experience may be hard to quantify or judge objectively, but in these reports, the judgments and decisions made by experienced crews were well reasoned, well timed, and effective.
- Time pressure, and related higher levels of workload, are factors in crew response.
  - More time pressure and workload prior to the malfunction may impact pilot response negatively.
- Additional observations related to
  - Flightcrew decisions about whether to operate the aircraft manually,
  - The potential impact of the flightcrews' English language skills on their interpretation of the situation,
  - The captain's technique in crosschecking right and left side flight deck displays, which was sometimes helpful, and
  - The roles and tasks that the different flightcrew members assumed during the event.
- Final thoughts
  - The types of unexpected situations observed in these accidents and ASRS events continue to occur.
  - While technological problems can be addressed in many cases, human behavior is harder to change
  - Pilot experience with unexpected situations in modern aircraft varies and will continue to evolve as the pilot workforce changes

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ENGINEERING ETHICS

MORALS AND ETHICS

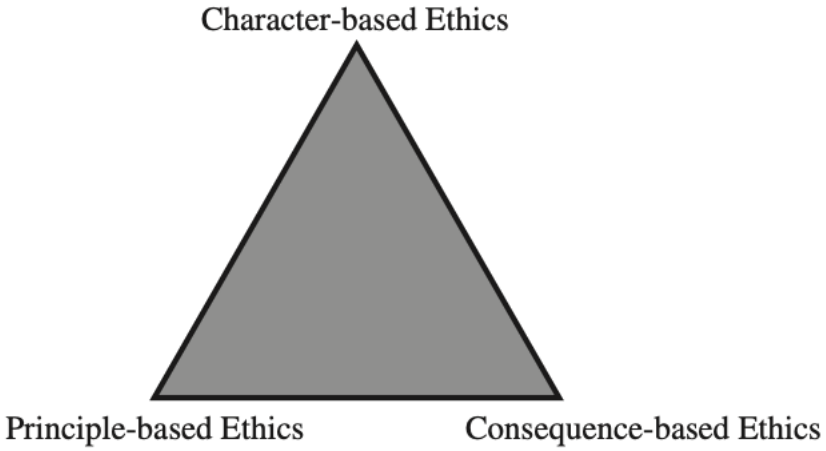
- "Morals are those rules we govern ourselves by, the principles we live by. Ethics is a field in philosophy that examines value judgments of right and wrong action."
  - "Ethics is the careful, philosophical study of morals…"
- Engineering is a "profession" – a *self-regulating occupation.*
- Ethics are applied to moral dilemmas in particular fields or professions, e.g.,
  - Inadequate review of engineering design work.
  - Pressure to falsify billable hours.
  - Performing work on which the engineer does not have competency.

From Chapter 1 of Starrett, Lara, & Bertha. (2017) *Engineering Ethics: Real World Case Studies.* ASCE Press (available under Canvas Reading List)

ETHICS AND LAWS

- "Ethics will tell you to drive only as fast as you can safely operate your vehicle; the law tells you that you can drive 35 mph on this road, and 45 mph on that one."

- You must know and apply both the law and ethical principals to guide professional conduct and to make choices when there are irreconcilable conflicts.



The Ethical Triangle.

From Chapter 1 of Starrett, Lara, & Bertha. (2017) *Engineering Ethics: Real World Case Studies.* ASCE Press (available under Canvas Reading List)

ETHICS CODES

- Professional Societies, e.g.,
  - Human Factors and Ergonomics Society (HFES)
  - National Society of Professional Engineers (NSPE)
  - American Institute of Aeronautics and Astronautics
  - Examples ethics codes online
    - https://www.hfes.org/Membership/Code-of-Ethics
    - https://www.nspe.org/career-growth/nspe-code-ethics-engineers
- Some focus areas
  - Data privacy
  - Institutional review boards (IRBs) – data from human subjects
  - Publication ethics
- Example Code of conduct
  - https://www.oge.gov/web/oge.nsf/resources_standards-of-conduct

From NSPE website:

I. Fundamental Canons

Engineers, in the fulfillment of their professional duties, shall:

1. Hold paramount the safety, health, and welfare of the public.
2. Perform services only in areas of their competence.
3. Issue public statements only in an objective and truthful manner.
4. Act for each employer or client as faithful agents or trustees.
5. Avoid deceptive acts.
6. Conduct themselves honorably, responsibly, ethically, and lawfully so as to enhance the honor, reputation, and usefulness of the profession.

OFFICE OF GOVERNMENT ETHICS (OGE.GOV)

"Public service is a public trust, requiring employees to place loyalty to the Constitution, the laws and ethical principles above private gain."

–Principle 1 of *The 14 Principles of Ethical Conduct for Employees of the Executive Branch*

- Other principles address conflict of interest, improper use of non-public information, receiving gifts, making unauthorized commitments on behalf of the government, outside employment, etc.

SOME ENGINEERING ETHICS CASES FOR STUDY

Challenger Space Shuttle

Deepwater Horizon Oil Spill

Citicorp Tower Design Flaw (NYC)

GM Ignition Switch

Flint (Michigan) Water Crisis

Volkswagen Emissions Scandal

Cambridge Analytica/Facebook

Boeing 737MAX Accidents

MIT College of Computing Case Studies

PERSONAL EXPERIENCES

- What kind of ethics training have you received?
- What do you recall from the training?
- Have you found yourself in situations where the training was helpful (or not)?

REFERENCES

- FAA AC 25.1309-1B (Aug 30, 2024). System Design and Analysis. (prior version A dated 6/21/1988)
- Chandra, D.C., Sparko, A.L. Kendra, A. (2023). Flightcrew Response to Aircraft System Failures, Malfunctions, and Systems Not Functioning as Expected. DOT-VNTSC-FAA-23-05. U.S. DOT Volpe National Transportation Systems Center. https://rosap.ntl.bts.gov/view/dot/72503
- Starrett, S.K. Lara, A.L., & Bertha, C. (2017) Engineering Ethics: Real World Case Studies. ASCE Press (available under Canvas Reading List)
- https://oge.gov/web/oge.nsf/resources_standards-of-conduct
- Jennings, Marianne M. and Trautman, Lawrence J., Ethical Culture and Legal Liability: The GM Switch Crisis and Lessons in Governance (December 14, 2015). 22 Boston University Journal of Science and Technology Law (2016), Available at SSRN: https://ssrn.com/abstract=2691536 or http://dx.doi.org/10.2139/ssrn.2691536